How Do Housing Market Responses to Macroeconomic Shocks Differ Across U.S. States?

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Introduction — What Are We Studying and Why?

The Problem

- Home ownership is the primary wealth-building vehicle for U.S. families
- CPI hit 40-year highs in 2021–2022, driven by COVID-era supply disruptions
- Home prices rose 40%+ nationally between 2020 and 2023 (FHFA, 2024)
- Fed raised rates from ≈ 0% to over 5% in just two years
- Mortgage affordability reached historic lows for first-time buyers

The Gap

- Most research treats the entire U.S. as one housing market
- National averages obscure major state-level variation
- California & New York: high-cost, supply-constrained
- Texas: elastic supply, lower price levels
- Arizona: rapid Sun Belt growth post-2020
- No study has tested cross-state heterogeneity in a panel framework with a COVID structural break

Statement of Problem

Existing research relies on national-level data and reports a single average estimate — masking how states with fundamentally different housing structures respond to the same macroeconomic shocks.

Supply Elasticity

Zoning and building restrictions vary widely across states. Inelastic supply amplifies the price impact of demand-side shocks like rate hikes (Glaeser & Gyourko, 2018).

Price Level Variation

A 1% rate increase has very different affordability consequences in CA/NY versus TX or OH. Pooled national models cannot capture this asymmetry.

Post-COVID Regime Change

The pandemic reshaped migration, remote work, and demand patterns. The pre-2020 relationship between macro shocks and housing may no longer hold.

Research Objectives

Q1

Baseline

How do inflation and interest rate changes affect housing activity at the national level?

Q2

State Heterogeneity

Do those effects differ across states with structurally different housing markets?

Q3

COVID Regime Shift

Did the macroeconomic–housing relationship change after Q1 2020?

Q4

Predictive Modeling

Can we forecast housing activity using inflation, interest rates, and state-level house prices?

Q2 is the primary research focus of this study

Literature Review

Case & Shiller (1989)	Housing markets are inherently local. Regional conditions — not national averages — drive price behavior. Early evidence that cross-region heterogeneity matters.
Glaeser & Gyourko (2018)	Zoning restrictions make housing supply inelastic in many cities. Supply-constrained markets amplify demand-side shocks from rate increases — directly motivates state comparisons.
Blanchard & Bernanke (2023)	U.S. pandemic inflation was both supply- and demand-driven. Each type may affect housing differently since housing is simultaneously a consumption good and an investment.
McCracken & Ng (2016)	FRED-MD/FRED-QD enables joint modeling of macroeconomic variables, revealing dynamics that single-variable methods miss. This study uses the Q1 2026 vintage.

Research gap: No study combines cross-state panel heterogeneity with a COVID-era structural break test.

Deliverables

01 Research Paper

25–35 page academic paper covering all four research questions, written in APA format with full statistical reporting.

02 Panel Dataset

Cleaned, documented dataset merging FRED-QD (March 2026 vintage) with FHFA state HPIs for 7 states, Q1 1975–Q4 2025.

03 Replication Code

Fully documented Python or R codebase — data cleaning, stationarity tests, all four models, diagnostics, and figures.

04 Visualizations

Time series plots, impulse response functions (8–12 quarter horizon), and side-by-side state comparison charts.

Explanation of Data

FRED-QD — March 2026 Vintage

- Quarterly, Q1 1959 – Q4 2025
- 245 macroeconomic variables

- **Key variables:**
  - CPIAUCSL — inflation (All Urban CPI)
  - FEDFUNDS — short-term interest rate
  - GS10 — 10-Year Treasury yield
  - HOUST — national housing starts
  - PERMIT — new housing permits
  - PAYEMS — total nonfarm payroll

FHFA State House Price Indices

CA	High-cost, supply-constrained
NY	High-cost, supply-constrained
TX	Lower-cost, elastic supply
FL	High-growth, strong in-migration
OH	Moderate-cost, Midwest market
IL	Midwest with large urban center
AZ	Sun Belt, rapid post-2020 growth

204 observations per state · Q1 1975 – Q4 2025

Statistical Analysis Plan

M1

Baseline OLS

Outcome: National housing activity

Inflation + interest rate + lagged DV.
Diagnostics: VIF (collinearity), Breusch-Godfrey (serial correlation).

M2

State Panel

Outcome: State × quarter

State fixed effects + state-level HPI + inflation×state and rate×state interactions.
Clustered standard errors.

M3

COVID Regime

Outcome: Structural break

COVID binary (Q1 2020+) + inflation×COVID + rate×COVID interactions.
Chow test for structural break.

M4

Predictive

Outcome: Forecasting RMSE

VAR with impulse response functions (8–12 quarters).
Ridge & LASSO vs. OLS on held-out 2020–2025 period.

All variables transformed per FRED-QD appendix codes · ADF stationarity tests applied before modeling

Project Timeline

Summer 2025

Wks 1–2

Data Prep

Clean & merge FRED-QD + FHFA into panel dataset

Summer 2025

Wk 3

EDA & Stationarity

Exploratory analysis; ADF tests on all variables

Summer 2025

Wks 4–5

Models 1 & 2

Baseline OLS + state heterogeneity panel regression

Fall 2025

Wk 6

Model 3 / Chow

COVID regime + structural break test

Fall 2025

Wks 7–8

Model 4

VAR impulse responses + Ridge/LASSO forecasting

Fall 2025

Wks 9–11

Writing

Full draft of research paper & visualizations

Spring 2026

Wk 12

Defense

Committee review, revisions, final submission & defense

References

Blanchard, O., & Bernanke, B. S. (2023). What caused the U.S. pandemic-era inflation? NBER Working Paper No. 31417.

Case, K. E., & Shiller, R. J. (1989). The efficiency of the market for single-family homes. American Economic Review, 79(1), 125–137.

Federal Housing Finance Agency. (2024). FHFA house price index: State-level data. https://www.fhfa.gov/DataTools/Downloads/Pages/House-Price-Index-Datasets.aspx

Glaeser, E. L., & Gyourko, J. (2018). The economic implications of housing supply. Journal of Economic Perspectives, 32(1), 3–30.

McCracken, M. W., & Ng, S. (2016). FRED-MD: A monthly database for macroeconomic research. Journal of Business and Economic Statistics, 34(4), 574–589.

Thank You

Questions & Discussion

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